

Project Development and Completion Tracker											
Project ID:	LYAIA6D	Class	LYAIA6	Percentage of Project Completion: 100%		Copyright Status:	NA	Technology Transfer:	NO		
Project Domain (Don't Add New)	Cybersecurity	Source of Problem Statement	Corporate	Map Sustainable Development Goal: (Don't add New)		Paper Publication Status:	Published	Achievements:	Type Your Answer_1, 2		
Enter Problem Statement	To develop a system that detects deepfake images using deep learning techniques. It analyzes facial features and artifacts to classify images as Real or Fake, ensuring authenticity and reducing misinformation.					GitHub Link:			Meeting Count		12
Project Development Student Team Members	Enrollment Number	Name of Students	Class	Contact Number	Email ID	Tech Skill 1	Tech Skill 2	Tech Skill 3	Global Certification 1	Global Certification 2	
	MITU22B1CS0093	Akshay Jadhav	LYAIA-6	9373723606	akshayjadhav2021@gmail.co	Deep Learning (CNN)	TensorFlow, Keras	Python, Model Training			
	MITU22B1CS0050	Om Mangate	LYAIA-6	7026090955	omastomom@gmail.com	Computer Vision	OpenCV	Image Processing, Face Detection			
	MITU22B1CS00750	Satyam Shinde	LYAIA-6	9529902376	satyamshinde56@gmail.com	Data Handling	Pandas, NumPy	Data Preprocessing, Augmentation			
	MITU22B1CS0043	Unmesh Kakuste	LYAIA-6	9579305872	uk21kakuste@gmail.com	Model Evaluation	Scikit-learn	Accuracy, Precision, Recall, F1-score			
Project Guide Industry Mentor	Prof. Sonali Deshpande										
						Technology Stack	1	2	3	4	5
Review-1	Review Date	Comment by Jury/Mentor	Review Link			Front-end (Client-side)	Gradio Web-based UI				
Review-2	1-5-2026					Back-end (Server-side)	Python				
Review-3	3-25-2026					Database Layer	OpenForensics Dataset				
						Development Tools	Jupyter Notebook, VS Code, Git, GitHub				
						Deployment and Infrastructure	Gradio / Localhost				
						Specialization Instrument	TensorFlow, Keras, OpenCV, Matplotlib, NumPy, Pandas				
Project Description: A web-based system to detect deepfake images and provide explainability so I can trust the authenticity of the media.											
Story 1:	As a user, I want to upload an image via a Gradio interface to check if it is real or fake.										
Story 2:	As a user, I want the system to use an ensemble of pre-trained CNN models for accurate classification.										
Story 3:	As a user, I want to see a Grad-CAM visualization highlighting the manipulated regions in the image.										
Task 1:	Setup development environment, Git repository, and download the OpenForensics dataset.										
Task 2:	Perform data preprocessing and image augmentation using OpenCV, NumPy, and Pandas.										
Task 3:	Fine-tune pre-trained CNN models (EfficientNetB1, ResNet50, Xception) using transfer learning.										
Task 4:	Implement ensemble learning using soft voting to combine model predictions.										
Task 5:	Integrate Grad-CAM to visualize important regions and explain model predictions.										
Task 6:	Develop a web-based Gradio interface for image upload and real-time result display.										
Task 7:											
Acceptance Criteria											
1	The system successfully loads and preprocesses images from the OpenForensics dataset.										
2	The ensemble model achieves high accuracy in classifying images as Real or Fake.										
3	The Grad-CAM visualization accurately highlights manipulated areas in fake images.										
4	The Gradio interface allows users to upload images and displays predictions and Grad-CAM in real-time.										

This is actual Development planning and tracking. Don't add documentation, survey, topic finalization, etc. activities.

Sprint No.	Task Name	Sub Tasks	Task Status	Assigned To	Assigned Date	Deadline	Start Date	Completion Date	Completion Status
Sprint 1									
1	Goal: Data planning and baseline setup		Complete	Akshay Jadhav	1/1/2026	1/5/2026	1/1/2026	1/5/2026	100%
2			Complete	Om Mangate	1/6/2026	1/10/2026	1/6/2026	1/10/2026	100%
3	Acquire OpenForensics dataset		Complete	Satyam Shinde	1/11/2026	1/15/2026	1/11/2026	1/15/2026	100%
4			Complete	Unmesh Kakuste	2026-01-16	2026-01-20	2026-01-16	2026-01-20	100%
5	Data augmentation		Complete	Akshay Jadhav	2026-01-21	2026-01-31	2026-01-21	2026-01-31	100%
Sprint 2									
	Goal: Core model training		Complete						
1	Fine-tune EfficientNetB1		Complete	Om Mangate	2026-02-01	2026-02-05	2026-02-01	2026-02-05	100%
2	Fine-tune ResNet50		Complete	Satyam Shinde	2026-02-06	2026-02-10	2026-02-06	2026-02-10	100%
3	Fine-tune Xception		Complete	Unmesh Kakuste	2026-02-11	2026-02-15	2026-02-11	2026-02-15	100%
4	Evaluate individual models		Complete	Akshay Jadhav	2026-02-16	2026-02-20	2026-02-16	2026-02-20	100%
5	Implement soft voting ensemble		Complete	Om Mangate	2026-02-21	2026-02-28	2026-02-21	2026-02-28	100%
Sprint 3									
	Goal: Nano-Banana-diffusion		Complete						
1	Research diffusion models		Complete	Satyam Shinde	2026-03-01	2026-03-05	2026-03-01	2026-03-05	100%
2	Integrate Nano-Banana		Complete	Unmesh Kakuste	2026-03-06	2026-03-10	2026-03-06	2026-03-10	100%
3	Adapt for deepfake detection		Complete	Akshay Jadhav	2026-03-11	2026-03-15	2026-03-11	2026-03-15	100%
4	Test adaptation performance		Complete	Om Mangate	2026-03-16	2026-03-20	2026-03-16	2026-03-20	100%
5	Refine model parameters		Complete	Satyam Shinde	2026-03-21	2026-03-31	2026-03-21	2026-03-31	100%
Sprint 4									
	Goal: Testing, tuning, and presentation		Complete						
1	Implement Grad-CAM		Complete	Unmesh Kakuste	2026-04-01	2026-04-05	2026-04-01	2026-04-05	100%
2	Develop Gradio Interface		Complete	Akshay Jadhav	2026-04-06	2026-04-10	2026-04-06	2026-04-10	100%
3	Final system testing		Complete	Om Mangate	2026-04-11	2026-04-15	2026-04-11	2026-04-15	100%
4	Tune hyperparameters		Complete	Satyam Shinde	2026-04-16	2026-04-20	2026-04-16	2026-04-20	100%
5	Prepare presentation		Complete	Unmesh Kakuste	2026-04-21	2026-04-30	2026-04-21	2026-04-30	100%

Publication Details									
Sr. No	Paper Title	Name of Journal	Year	Authors	URL	DOI	Volume	Page no.	Publisher
1			2026						
2									
Patent Details									
Sr. No.	Title	Inventors	Application No.	Patent Number	Filing Country	Subject Category	Filing Date	Publication Date	Publication Status
1									
Copyright Details									
Sr. No.	Title of work	Name of Applicants	Registration No.	Dairy Number	Date	Status			
1									
Event and Participations Details									
Sr. No.	Name of Event	Type of Event	Date	Type of Participation	Details of Prize won				
1									
2									

Add Weekly Meetings Details (Minimum 12 meetings)

Meeting No.	Date	Attendees	Topic	Action Items	Assigned to	Status
1	2/6/2026	Akshay, Om, Satyam, Unmesh	Baseline Training Setup	Set up EfficientNet baseline	Om, Akshay	✓
2	2/13/2026	Akshay, Om, Satyam, Unmesh	Xception & ResNet Training	Train ResNet and Xception models	Satyam, Unmesh	✓
3	2/20/2026	Akshay, Om, Satyam, Unmesh	del Evaluation & Checkpoi	Evaluate individual models	Akshay	✓
4	2/27/2026	Akshay, Om, Satyam, Unmesh	Int 2 Review & Diffusion P	Implement soft voting ensemble	Om, Satyam	✓
5	3/6/2026	Akshay, Om, Satyam, Unmesh	Diffusion Dataset Integratio	Research and integrate Nano-Banana	Satyam, Unmesh	✓
6	2026-03-13	Akshay, Om, Satyam, Unmesh	ne-Tuning on Diffusion Dat	Adapt models for deepfake detection	Akshay, Om	✓
7	3/20/2026	Akshay, Om, Satyam, Unmesh	Generalization Gap Analysis	Test adaptation performance	Om, Satyam	✓
8	3/27/2026	Akshay, Om, Satyam, Unmesh	3 Review & Checkpoint Se	Refine model parameters	Satyam, Unmesh	✓
9	4/3/2026	Akshay, Om, Satyam, Unmesh	Final Testing & Validation	Validate ensemble on test set	Akshay, Om	✓
10	4/10/2026	Akshay, Om, Satyam, Unmesh	Grad-CAM & Explainability	Implement Grad-CAM visualization	Unmesh, Akshay	✓
11	4/17/2026	Akshay, Om, Satyam, Unmesh	Radio Demo UI Developme	Develop web-based Gradio interface	Akshay, Satyam	✓
12	4/24/2026	Akshay, Om, Satyam, Unmesh	Final Report & Viva Prep	Prepare final presentation	Unmesh, Om	✓